Analysis of the U.S. Domestic Air Network (2025)

Daniel Marren

Github Repository: https://github.com/danny-m-22/Airport-Network-Analysis
Interactive map: https://danny-m-22.github.io/Airport-Network-Analysis/

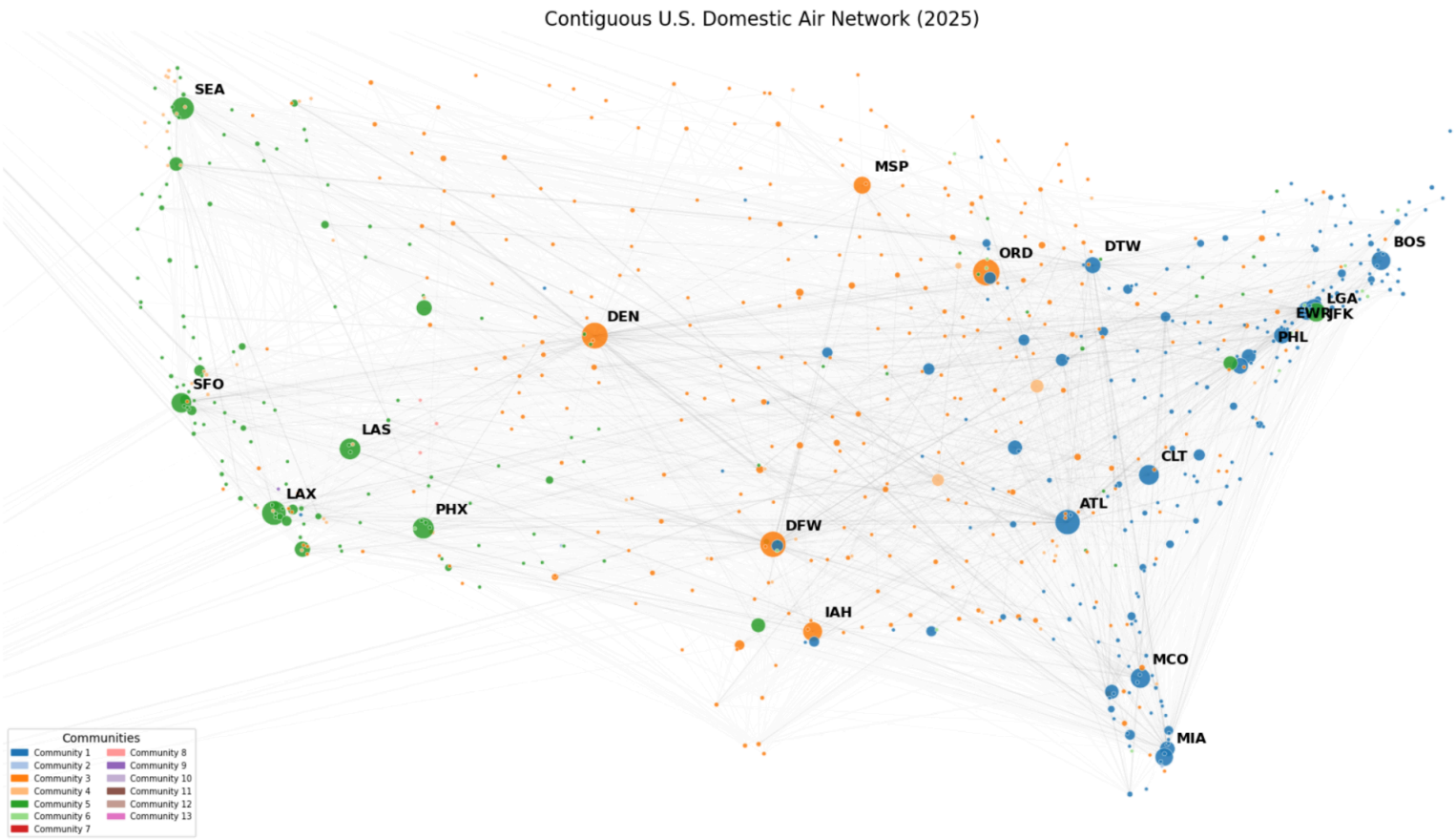

Figure 1. U.S. domestic air network (contiguous states). Node color indicates Louvain community assignment, node size reflects total annual segment passengers, and edge thickness and opacity reflect passenger volume on each route. Top 20 airports by passenger volume are labeled.

- The U.S. domestic air network is a critical infrastructure system, and its reliability depends on a small number of highly connected hub airports. Understanding the properties of this network, including hub concentration, regional community structure, and geographic bottlenecks, has direct implications for resilience planning. Disruptions at high-connectivity nodes like ORD or DFW propagate disproportionately across the system.
- This analysis provides a data-driven foundation for identifying which airports are indispensable, which can help inform decisions around investment and contingency routing and can help understand the downstream effects of weather, operational, or policy-driven disruptions.

- The airport system comprises 1,203 airports and 19,001 routes, and this forms a single connected component after filtering to valid IATA codes — i.e., valid three-letter airport codes — and non-zero passenger segments.
- Louvain community detection identified 13 communities, which mostly corresponds to geographic regions. Three large communities are particularly dominant: a western cluster (green) anchored by LAX, SFO, SEA, LAS, and PHX, a central cluster (orange) anchored by DFW, DEN, and ORD, and an eastern cluster (blue) anchored by ATL, CLT, and Northeastern airports.
- The network exhibits strong hub concentration: the top 5 airports (ATL, ORD, DFW, DEN, LAX) collectively handle 18.9% of all domestic segment passengers, and the top 10 handle 31.5%; the top 20 airports handle 51%.
- Route connectivity analysis reveals that ORD is the most connected airport with approximately 300 unique domestic routes, followed by DEN and MSP. Interestingly, ANC ranks 4th in route connectivity, which shows Alaska's dependence on air travel for inter-state access. Smaller hubs such as SLC, BNA, and AUS appear in the top 20.

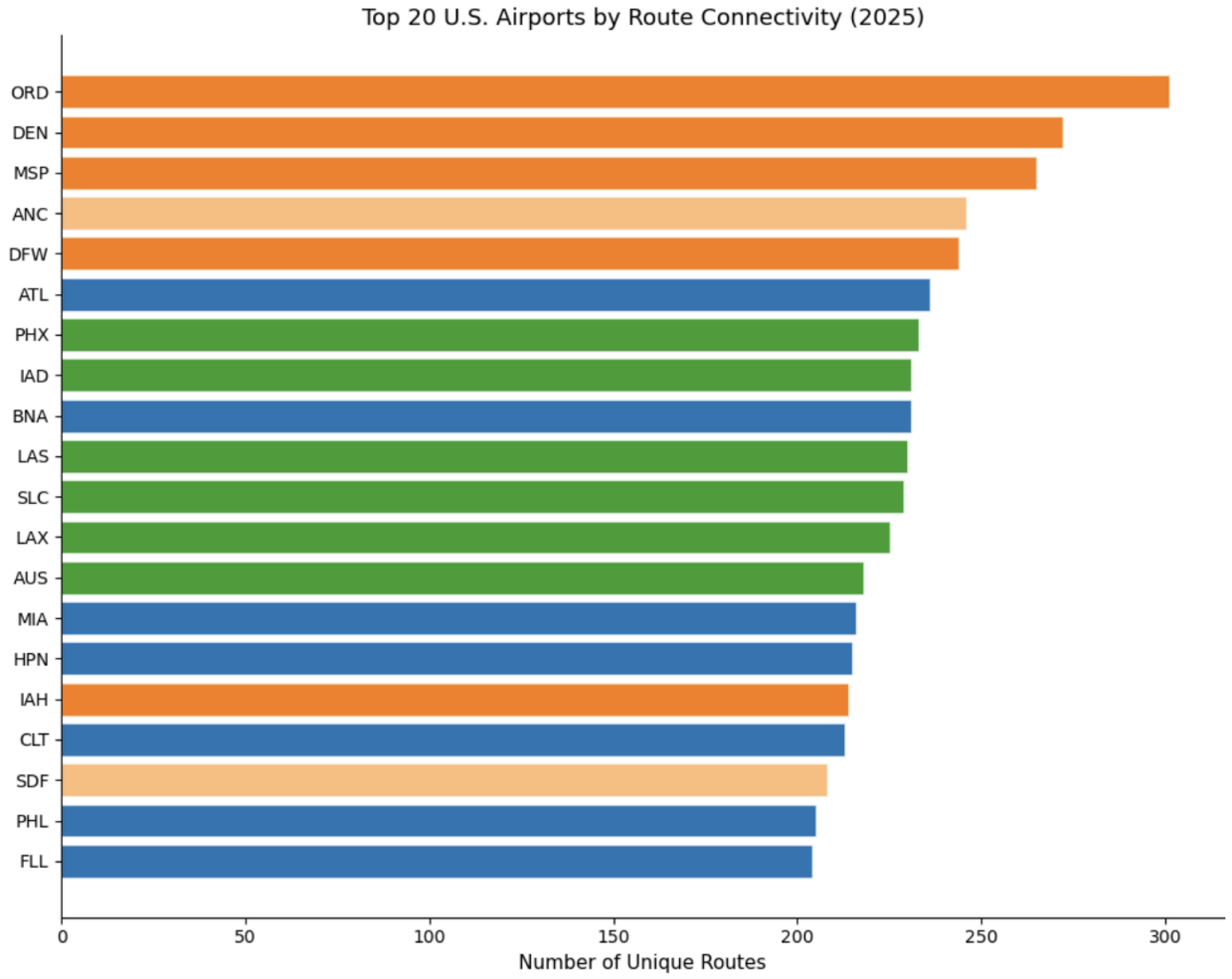

Figure 2. Top 20 U.S. airports by number of unique domestic routes (2025). Node color matches community assignment from Figure 1. ORD, DEN, and MSP lead in connectivity; ANC's high rank reflects how isolated Alaska is and how it has its own airport sub-network.

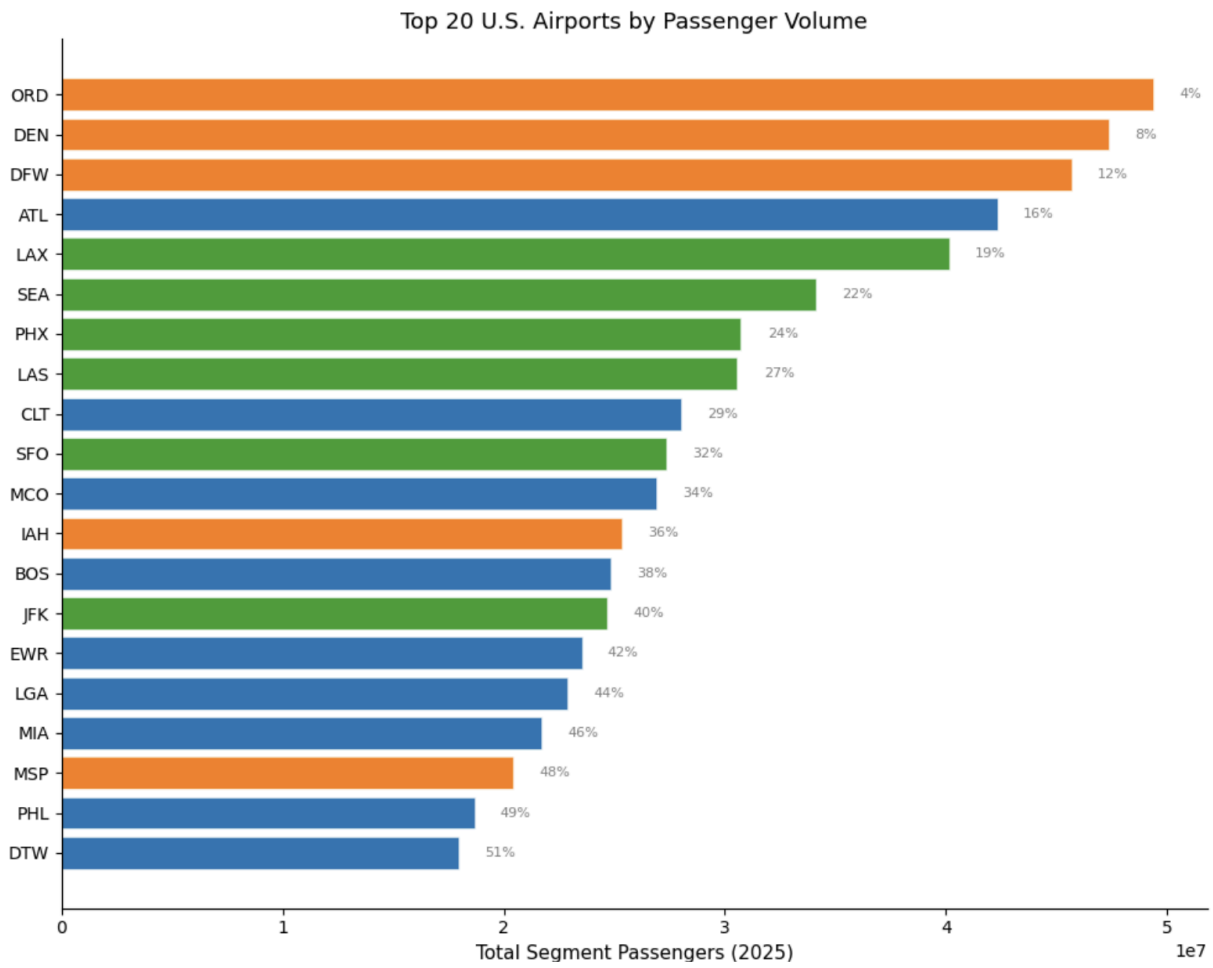

Figure 3. Top 20 U.S. airports by total segment passengers (2025). Cumulative percentage labels indicate share of total domestic passenger activity. Node color matches community assignment from Figure 1.

- Route and passenger data were sourced from the Bureau of Transportation Statistics (BTS) T-100 Domestic Segment database (DB28), and it covers January through December 2025. Data were filtered to routes with valid three-letter IATA airport codes and non-zero passenger counts, then aggregated to the route level with total annual passengers as edge weights.
- Airport geographic coordinates were obtained from OpenFlights. A weighted undirected graph was constructed using NetworkX, and community structure was identified via the Louvain algorithm.
- A limitation of this analysis is that passenger counts reflect enplaned segment passengers and not true origin-to-destination volumes. A connecting passenger is counted once per segment, inflating hub airport totals relative to true end-to-end demand. Relative hub rankings and network structure remain valid, though.